STATISTICAL LEARNING • ADVANCED SELF-STUDY

LINEAR METHODS FOR CLASSIFICATION

ESL • Chapter 4 | From geometric boundaries to likelihood, Fisher directions, and margins

How to use this lecture

Part I teaches the mathematics from intuition through derivations, with original graphs, numerical examples and checkpoints. Part II turns the ideas into small, reproducible R commands, with comments explaining what each line is doing. Stop after a checkpoint, solve it, then compare with the answer key.

> **PRACTICE RULE**
>
> Read → work the example by hand → run the R block → verify the output → interpret what changed. The code deliberately avoids loops, custom pipelines, and complex helper functions.

Companion files: a standalone R script and the user-supplied chapter PDF. All teaching text and diagrams in this lecture are original explanations, not reproduced chapter pages.

The reading map

This lecture follows the supplied **ESL Chapter 4** (printed pages 101–135): §4.1 classification geometry; §4.2 indicator-matrix regression; §4.3 LDA, regularization, computation, and reduced rank; §4.4 logistic and multinomial logistic regression; and §4.5 separating hyperplanes. Extra teaching examples extend the derivations and connect the theory to reproducible R.

> **TWO COMPLEMENTARY QUESTIONS**
>
> **Statistical:** What probability or density model describes an event? **Geometric:** What surface divides the feature space? Logistic regression and LDA often produce the same geometric type of boundary while making different statistical assumptions.

Prerequisites you will build: conditional probability, Bayes' theorem, vectors, dot products, matrix multiplication, covariance, maximum likelihood, derivatives, and optimization. Each appears when it becomes useful; mastery of advanced linear algebra is not assumed.

PART I — THEORY: PRIMER TO ADVANCED

1. The classification problem and geometry

Chapter anchor: *ESL §4.1, printed pp. 101–103.*

In regression, the target is quantitative, such as birth weight in grams. In classification, the target is a label: low vs normal birth weight, a diagnosis category, or one of K disease subtypes. The mathematical goal is usually to learn the conditional probability P(Y=k | X=x), then choose a label by an explicit decision rule.

Let x=(x1,…,xp) be a predictor vector. A **score** for class k is a function δk(x). We predict the class with the largest score. A **decision boundary** between classes a and b contains the points where δa(x)=δb(x). If each δk is linear in x, boundaries are hyperplanes; in two dimensions they are lines.

$$\delta_k(x) = \beta_{k0} + \beta_k^\top x, \qquad \hat{y} = \arg\max_k \delta_k(x)$$

> **WORKED EXAMPLE**
>
> **Question.** A classifier has scores δ0(x)=0.6+x1 and δ1(x)=−0.2+2x2. Where is the boundary?
>
> **Solution.** Set the scores equal: 0.6+x1=−0.2+2x2, so x1−2x2+0.8=0. That is a straight line. On one side δ0>δ1; on the other δ1>δ0. Three dimensions would yield a plane, not a fundamentally different method.

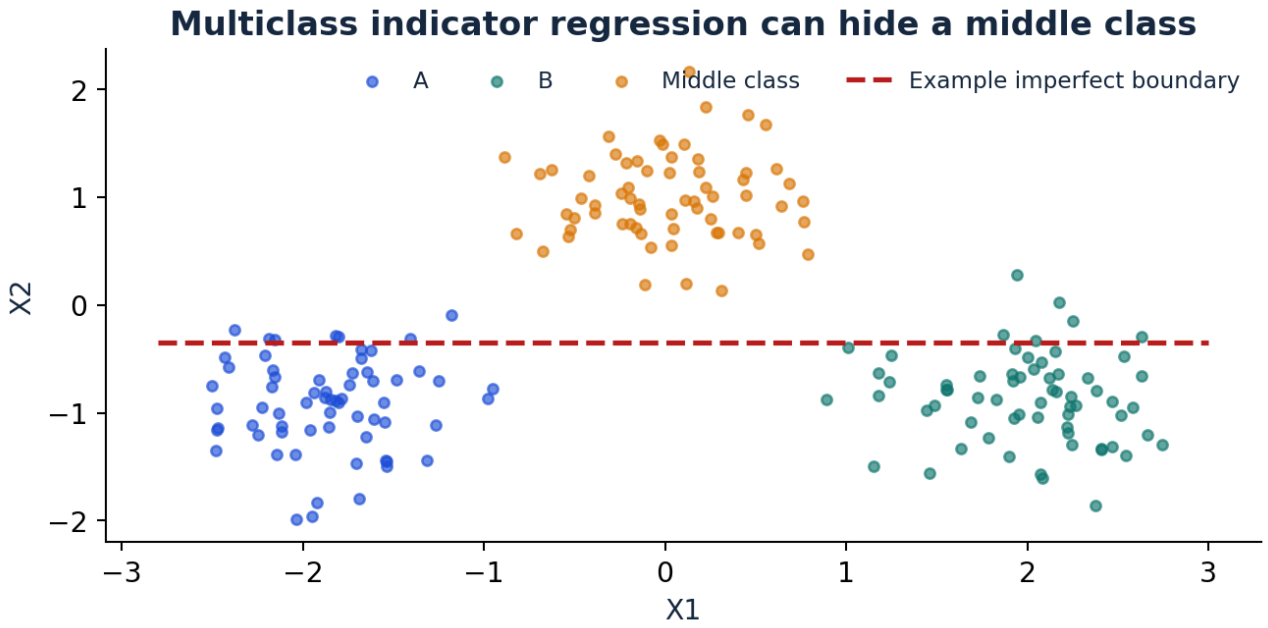


Figure. *Classes and boundaries in two-dimensional feature space. A linear rule may be inadequate when one class lies between others, motivating transformations or another model.*

> **MASTERY CHECK 1**
>
> If a linear decision score is β0+β1x1+β2x2, what does β=(0,0) mean for the geometry? Why does the intercept change the location but not the orientation of a boundary?

2. The probability primer: conditional probability, Bayes, and costs

A predictive model outputs probabilities *conditional on measured predictors*; it does not by itself establish causes. Bayes' theorem separates three ingredients: (i) prevalence or prior class probability πk, (ii) how likely the observed features are under each class, fk(x), and (iii) posterior class probability.

$$P(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{\ell=1}^{K} \pi_\ell f_\ell(x)}$$

> **WORKED EXAMPLE**
>
> **Question.** Prevalence of an outcome is 20%. A biomarker value has density 0.30 among cases and 0.10 among noncases. Find P(case | biomarker).
>
> **Solution.** The numerator is 0.20×0.30=0.06. The denominator is 0.06+0.80×0.10=0.14. The posterior is 0.06/0.14≈0.429. A relatively case-enriched biomarker value does not imply a posterior greater than 50% when prevalence is low.

The Bayes classifier under equal misclassification losses chooses the class with the largest posterior probability. Under unequal losses, the threshold changes. With false negatives four times as costly as false positives, the event threshold is 1/(1+4)=0.20, assuming well-calibrated probabilities and those are the only two error costs.

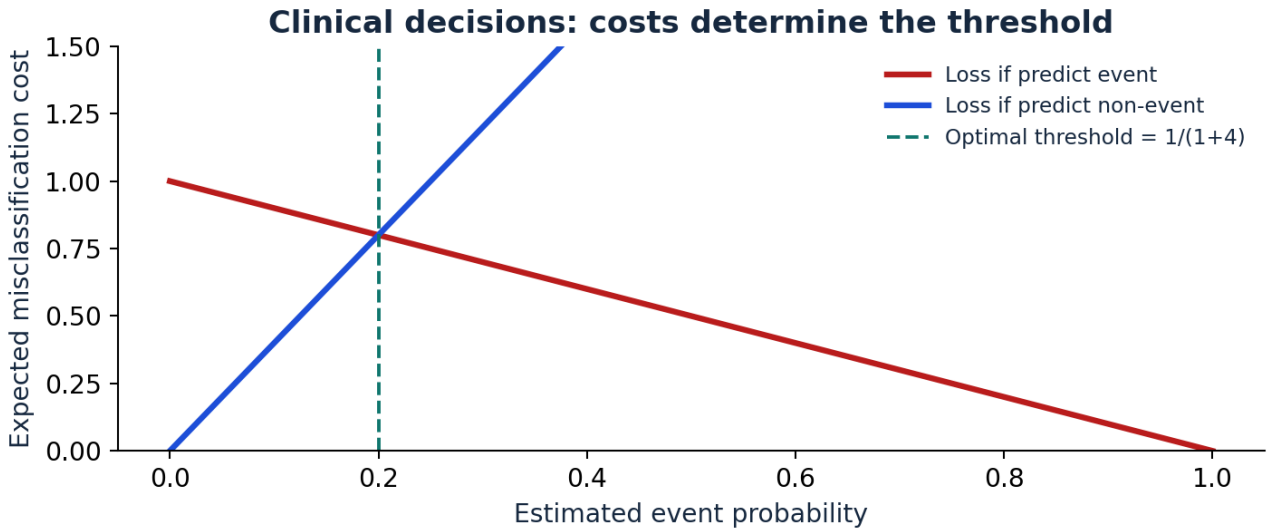


Figure. *Expected decision costs shift the optimal cutoff; the default 0.50 threshold has no universal clinical justification.*

> **MASTERY CHECK 2**
>
> If $\pi 1$ falls while both f1(x) and f0(x) stay fixed, how must the posterior probability of class 1 change? Show the change using Bayes' formula.

3. Why linear regression of indicators is tempting — and fragile

Chapter anchor: *ESL §4.2, pp. 103–106; see Fig. 4.2 and its middle-class masking example.*

For K classes we may encode each observation as a row of a one-hot indicator matrix Y of dimension n×K. Ordinary multivariate least squares estimates $B=(X^TX)^{-1}X^TY$. The fitted K-dimensional vector can be used to select the largest component. With just two labels, this becomes the linear probability model.

$$\hat{B} = (X^\top X)^{-1} X^\top Y, \qquad \hat{G}(x) = \arg\max_k [\tilde{x}^\top \hat{B}]_k$$

The fitted components **are not automatically valid probabilities**: they can be negative, exceed one, or create masking in multiclass settings. A high training fit is not proof of a useful classification boundary. Expanding x with polynomial features makes the score linear in the *expanded* space but nonlinear in the original variables.

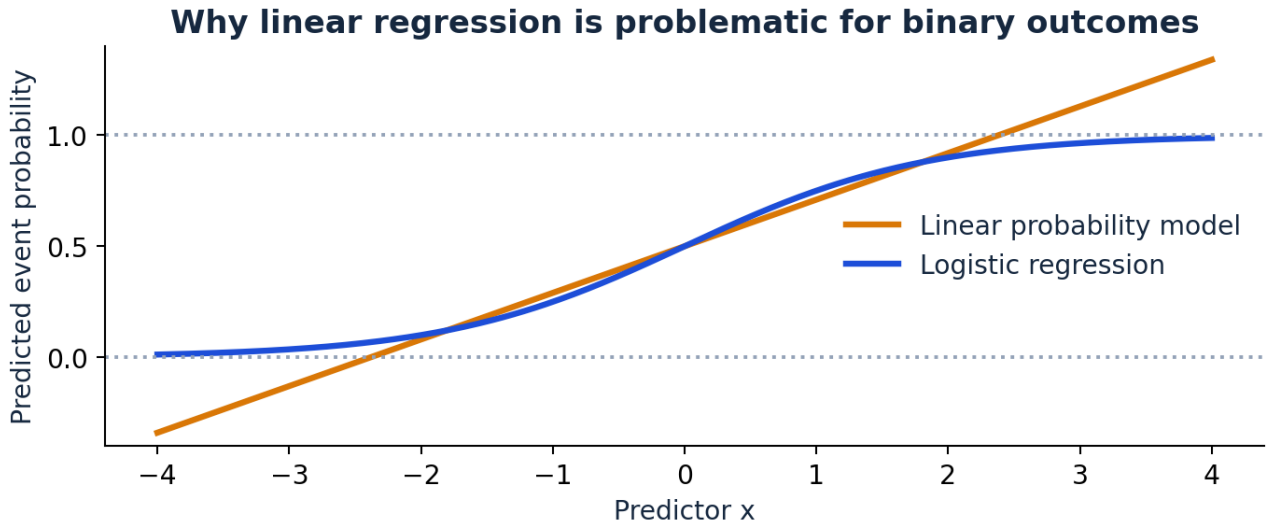


Figure. *Ordinary linear regression can predict probabilities outside [0,1], while the logistic link stays bounded.*

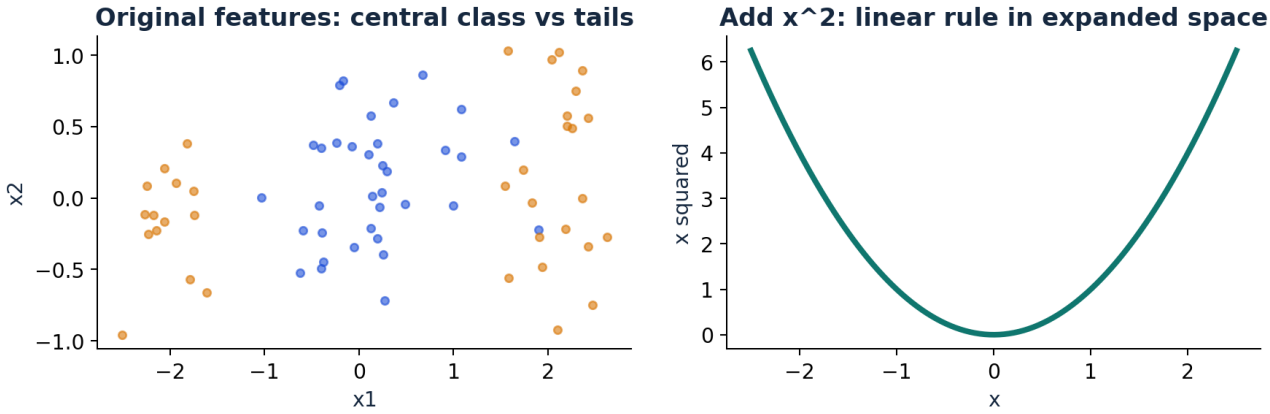


Figure. *A linear rule in (x, x^2) can create a curved rule in x. The geometry changes through feature engineering, not through a non-linear coefficient model.*

WORKED EXAMPLE

Question. For $\eta(x)=0.5+0.21x$, calculate the "probability" at $x=4$ and $x=-4$.

Solution. At 4, $\eta=1.34$, greater than one; at -4, $\eta=-0.34$, below zero. Least squares does not enforce the probability axioms. The numerical values are mathematical scores, not credible probability estimates.

MASTERY CHECK 3

Could a polynomial indicator regression create a curved decision boundary? If yes, why does this not solve all masking problems?

4. Odds, log-odds, and the logistic link

Chapter anchor: *ESL §4.4, printed pp. 119–125.*

For a Bernoulli outcome, p(x)=P(Y=1|x). Odds are p/(1−p); the logit is log[p/(1−p)]. The logistic model assumes that **log-odds, not probability itself**, are a linear function of predictors. Exponentiating a coefficient gives an odds ratio for a one-unit predictor change, holding other predictors fixed.

$$\eta = \beta_0 + \beta^\top x, \qquad p(x) = \frac{e^\eta}{1 + e^\eta}$$

$$\log \frac{p(x)}{1 - p(x)} = \eta, \qquad OR_j = e^{\beta_j}$$

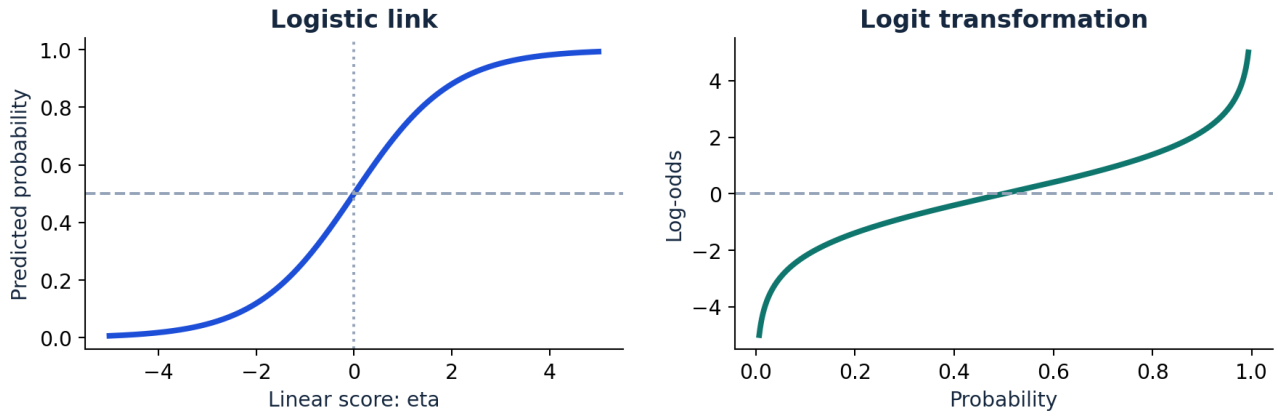


Figure. *The logistic link maps any real-valued linear score to a probability; the inverse logit turns probability into an unbounded real number.*

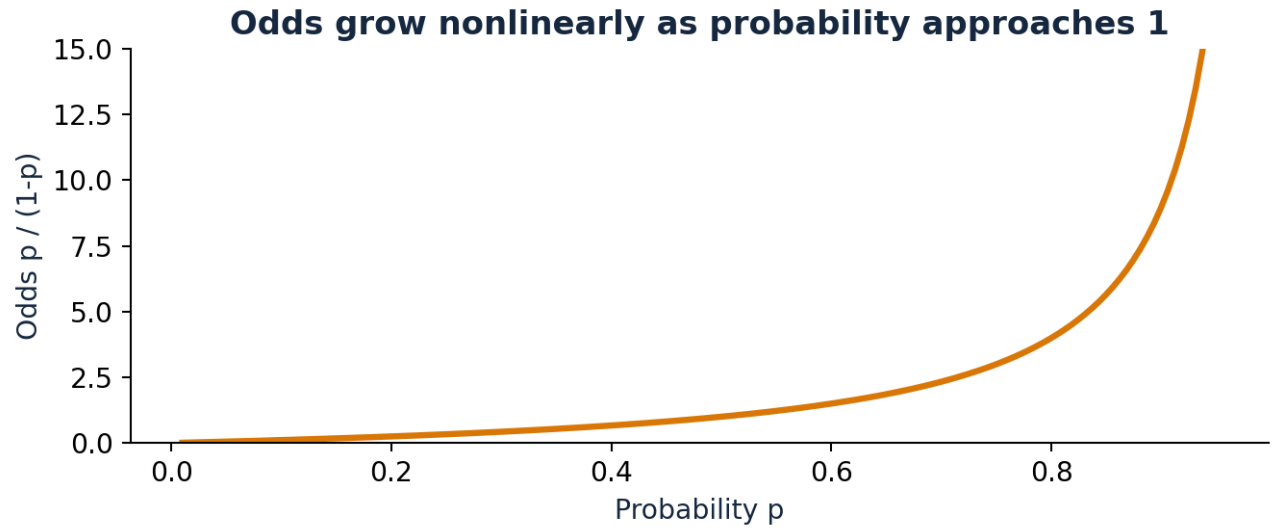


Figure. *Odds and probability should never be used interchangeably, especially for common outcomes.*

WORKED EXAMPLE

Question. Baseline event probability is 0.20 and a binary predictor has OR=1.50. Find the exposed probability.

Solution. Baseline odds=0.20/0.80=0.25; multiply by 1.50 to get 0.375; convert back: 0.375/1.375≈0.273. Risk rises from 20.0% to 27.3% — not to 30%, and not by 50 percentage points.

MASTERY CHECK 4

If β1=−0.7, what is exp(β1)? Does β1 mean a −70 percentage-point change in risk?

5. Maximum likelihood: what is actually optimized?

For independent Bernoulli outcomes $y_i \in \{0,1\}$ with modeled probabilities p_i, each observation contributes p_i if $y_i=1$ and $(1-p_i)$ otherwise. Multiplying over independent observations gives the likelihood; summing log contributions yields a more tractable objective.

$$\ell(\beta) = \sum_{i=1}^{n} \{y_i \log p_i + (1-y_i)\log(1-p_i)\}$$

$$\nabla \ell(\beta) = X^\top (y-p), \qquad H(\beta) = -X^\top W X$$

Here X contains the intercept and predictors; W is diagonal, with entries $p_i(1-p_i)$. The score $X^T(y-p)$ compares observed with modeled events across every predictor. The negative Hessian X^TWX measures local curvature. Fitting logistic regression is an **optimization** problem; in general there is no single closed-form least-squares answer.

WORKED EXAMPLE

Question. Start an intercept-only logistic model for five outcomes (0,0,0,1,1) at β0=0. Compute one Newton step.

Solution. At β0=0 every $p_i=0.5$. Score=$\Sigma(y_i-p_i)=2-2.5=-0.5$. Information=$\Sigma p_i(1-p_i)=5\times0.25=1.25$. Newton update is βnew=0+(−0.5/1.25)=−0.4, implying p≈0.401. The MLE would approach the observed event fraction 2/5=0.4.

6. Newton-Raphson as iteratively reweighted least squares

Chapter anchor: *ESL §4.4.1, printed pp. 120–121.*

The maximum-likelihood fitting algorithm can be written as a Newton iteration. Define $W=\text{diag}(p_i(1-p_i))$, and the working response $z=X\beta+W^{-1}(y-p)$. The next β solves weighted least squares: regression of z on X with weights W. This is IRLS. You do not need to program the algorithm to understand what `glm(..., family=binomial)` is solving.

$$\beta_{new} = \beta_{old} + (X^\top W X)^{-1} X^\top (y - p)$$

Two practical failures matter. **Complete separation** occurs when a linear rule perfectly separates observed classes and ordinary logistic coefficients can diverge. **Overfitting** occurs when apparent model fit is optimistic relative to new observations. Penalization or better validation targets the second issue; regularization can also stabilize the first, depending on method.

> **INFERENCE CAUTION**
>
> A Wald p-value relies on large-sample approximations and the fitted model. If variables are selected using the same outcomes, ordinary post-selection p-values are generally not valid without adjustment. Statistical significance is not predictive utility.

> **MASTERY CHECK 5**
>
> Why does W assign the largest weights near p=0.5, and very small weights when p≈0 or 1? Explain using Bernoulli variance p(1−p).

7. LDA from Gaussian likelihoods

Chapter anchor: *ESL §4.3, pp. 106–111.*

Linear discriminant analysis is **generative**: it models the distribution of X given each class, then uses Bayes' theorem. Assume X|Y=k follows a p-dimensional normal distribution with mean μk and a covariance matrix Σ shared across classes. Taking log posterior odds cancels the quadratic $x^T\Sigma^{-1}x$ terms; the remaining score is linear in x.

$$\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \tfrac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k$$

The classifier estimates priors πk, class means μk, and pooled within-class covariance Σ. The inverse covariance adjusts for feature scale and correlations: a difference along a high-variance direction carries less evidence than the same difference along a tightly concentrated direction.

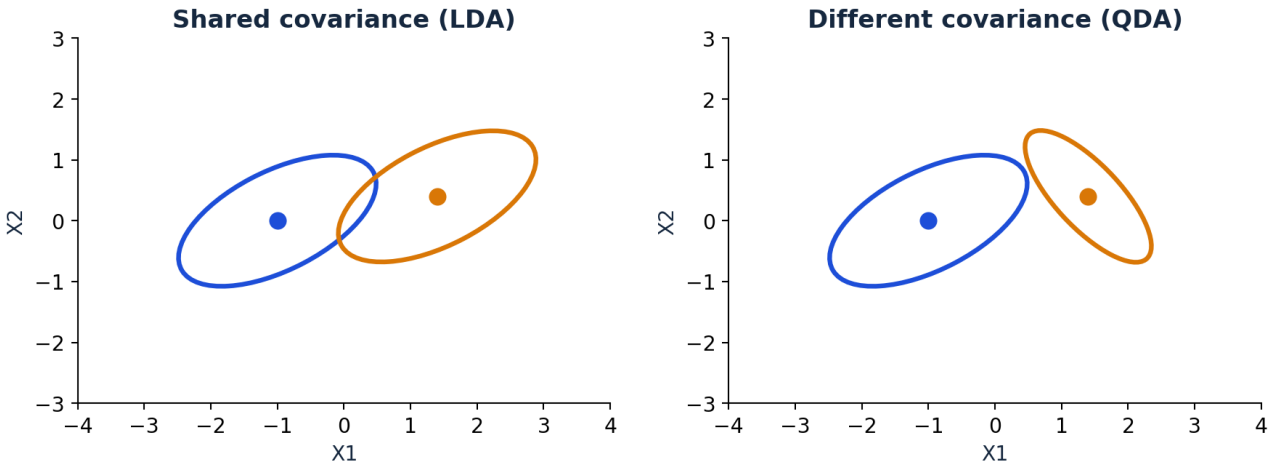


Figure. *LDA assumes a shared within-class covariance: contours can differ by their centers but not their shape and orientation. QDA allows those shapes to differ.*

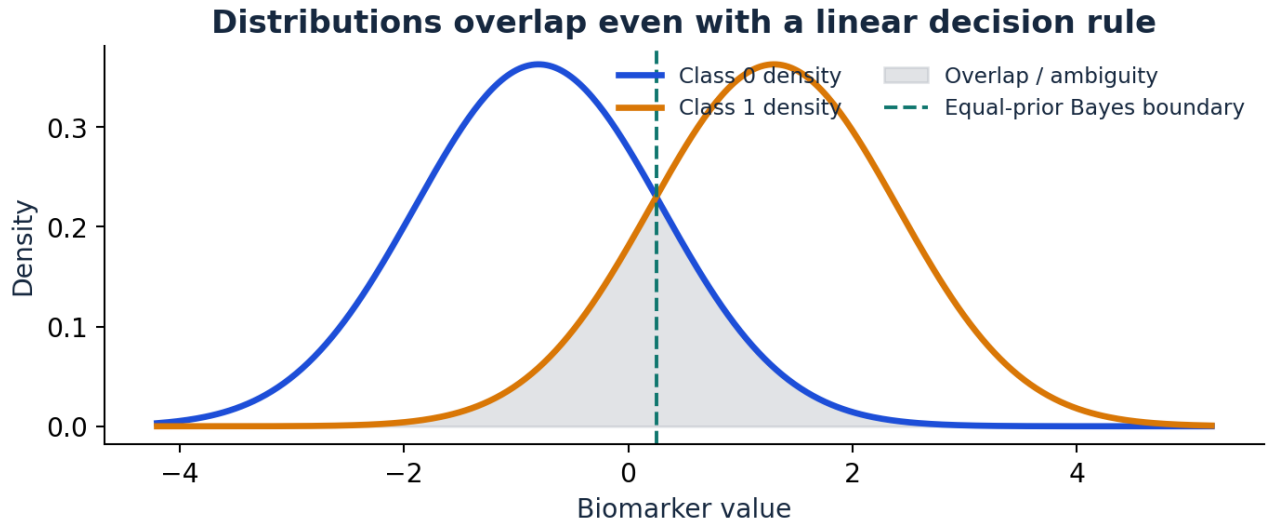


Figure. *The fitted Gaussian density ratio, together with prior probabilities, determines which class wins at each predictor value.*

WORKED EXAMPLE

Question. For a single predictor, X|0~N(0,1), X|1~N(2,1), and equal priors, find the Bayes boundary.

Solution. The log density ratio is $2x-2$. Predict class 1 when $2x-2>0$, so the boundary is $x=1$ (the midpoint of means). If the class-1 prior is 0.20, add $\log(0.20/0.80)=-1.386$: class 1 is selected only when $x>1.693$.

MASTERY CHECK 6

Why does replacing shared Σ with class-specific Σ_k prevent cancellation of the $x^T\Sigma_k^{-1}x$ terms? What geometric change follows?

8. QDA and regularized discriminant analysis

Chapter anchor: *ESL §4.3 and §4.3.1, pp. 110–113.*

Quadratic discriminant analysis assigns each class its own covariance matrix Σk. Because the quadratic terms no longer cancel between classes, QDA can form curved decision boundaries. It is more flexible but estimates more parameters. With p predictors, each unconstrained class covariance involves p(p+1)/2 distinct terms: a severe burden when sample sizes are limited.

$$\delta_k(x) = -\tfrac{1}{2}\log|\Sigma_k| - \tfrac{1}{2}(x-\mu_k)^\top \Sigma_k^{-1}(x-\mu_k) + \log \pi_k$$

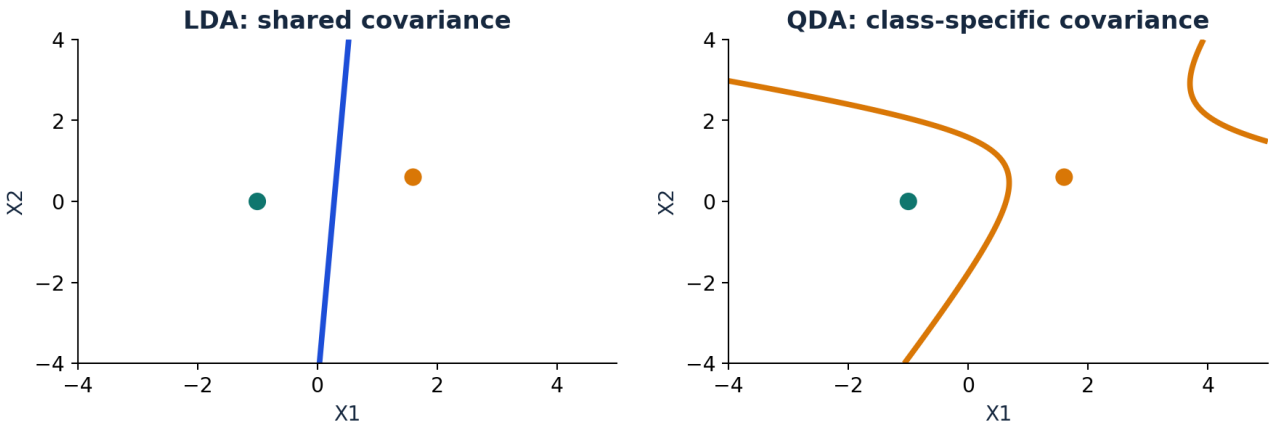


Figure. *In an illustrative two-class problem, common covariance produces a straight line and different covariance matrices produce curvature.*

Regularized discriminant analysis blends the class-specific Σk with the pooled Σ. In ESL notation, the mixing weight α controls how far we move from LDA (α=0) toward QDA (α=1). Estimate the tuning parameter using held-out data or cross-validation, never by inspecting the final test-set performance.

$$\Sigma_k(\alpha) = \alpha\Sigma_k + (1-\alpha)\Sigma, \qquad 0 \le \alpha \le 1$$

WORKED EXAMPLE

Question. Two classes have within-class variance estimates 1 and 9, while the pooled variance is 5. For α=0.25, find the regularized variances.

Solution. For class 1: 0.25×1+0.75×5=4. For class 2: 0.25×9+0.75×5=6. The highly variable variance estimates are drawn toward a common value, trading flexibility for stability.

9. Fisher's discriminant: the geometry of separation

Chapter anchor: *ESL §4.3.3, printed pp. 113–118.*

Fisher asks a geometric question without initially requiring Gaussian densities: which projection direction w makes class means far apart while keeping within-class spread small? Let B be between-class scatter and W within-class scatter. The Rayleigh quotient measures the ratio. For two classes, the maximizing direction is proportional to $W^{-1}(\mu 1 - \mu 0)$, when W is invertible.

$$J(w) = \frac{w^\top B w}{w^\top W w}, \qquad w \propto W^{-1}(\mu_1 - \mu_0)$$

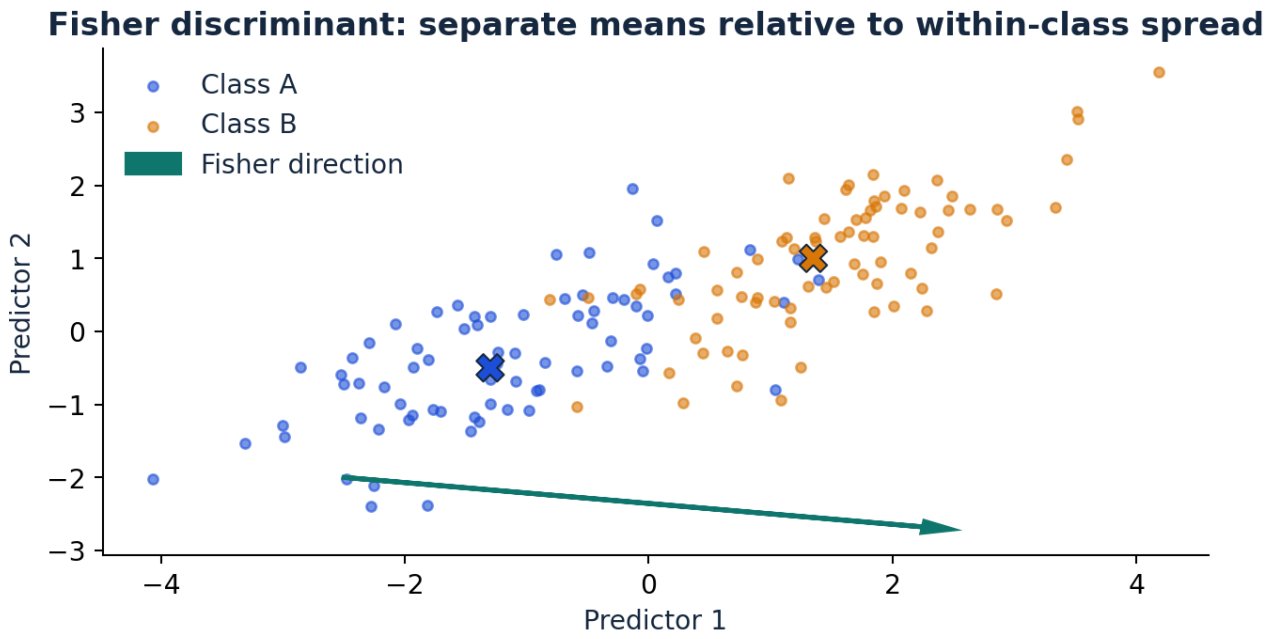


Figure. *Fisher chooses a discriminant axis informed by covariance, not merely the Euclidean line between centroids.*

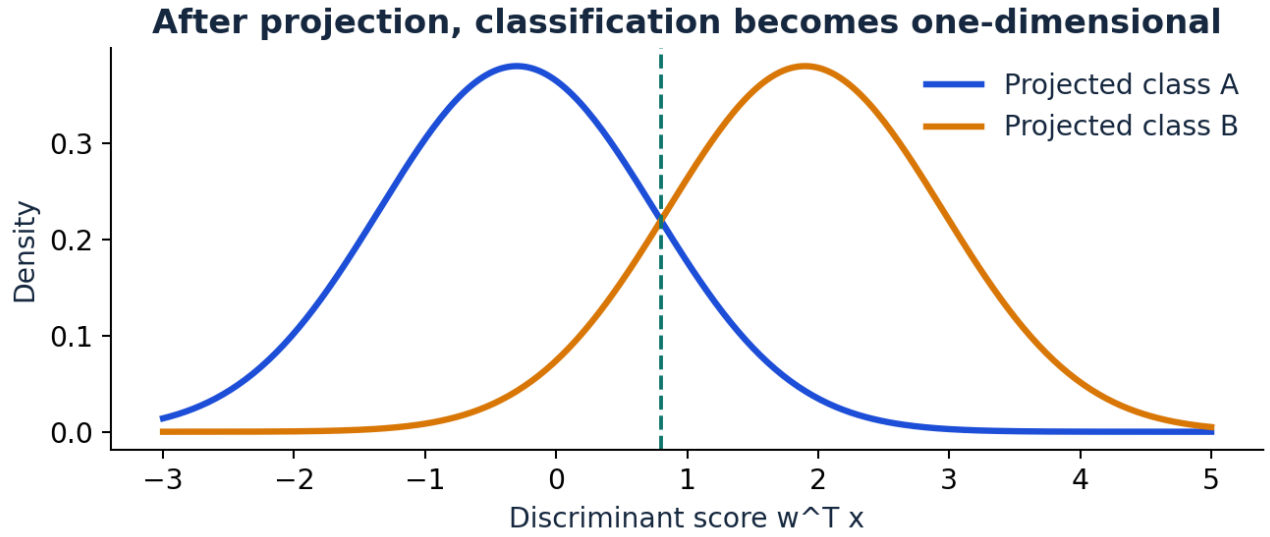


Figure. *Once projected, class probabilities or thresholds can be interpreted in a one-dimensional score.*

Reduced-rank LDA: With K classes, centered class means span at most K−1 dimensions. This explains why Fisher discriminants require at most min(p,K−1) nonzero directions. This is supervised dimension reduction: class labels determine what variation is useful, unlike ordinary PCA.

> **WORKED EXAMPLE**
>
> **Question.** The class means differ by μ1−μ0=(2,0). The within-class covariance is diag(4,1). Find the Fisher direction.
>
> **Solution.** W^{-1}=diag(1/4,1). Therefore w=(1/2,0), up to arbitrary scale. The first feature remains relevant, but its effect is discounted by its large within-class variance. For correlated features the inverse covariance also changes the direction of w.

> **MASTERY CHECK 7**
>
> Why can an 11-class problem have at most 10 nonzero Fisher canonical discriminants? Does having 500 features change this bound?

10. Multinomial logistic regression and the softmax

Chapter anchor: *ESL §4.4, printed pp. 119–120.*

With K>2 labels, choose a reference class K. For every non-reference class k, model the log of the posterior odds relative to K as a linear function of x. The softmax transformation converts scores into nonnegative class probabilities summing to one. Changing the reference class changes coefficient interpretations but not the underlying class probabilities when the model is fitted equivalently.

$$P(Y=k \mid x)=\frac{\exp(\eta_k(x))}{1+\sum_{j=1}^{K-1}\exp(\eta_j(x))}, \quad k<K$$

$$P(Y=K \mid x)=\frac{1}{1+\sum_{j=1}^{K-1}\exp(\eta_j(x))}$$

> **WORKED EXAMPLE**
>
> **Question.** Three classes have non-reference linear scores η1=log(2), η2=log(3); class 3 is the reference. Find all posteriors.
>
> **Solution.** Denominator=1+2+3=6, so probabilities are P1=2/6, P2=3/6, and P3=1/6. The three probabilities sum to one, which arbitrary one-vs-rest linear regressions do not guarantee.

> **MASTERY CHECK 8**

If you increase only η1, must the probability of class 2 stay unchanged? Explain the shared denominator.

11. L1 penalization: why sparse logistic models work

Chapter anchor: *ESL §4.4.4, printed pp. 125–127.*

Lasso logistic regression maximizes log-likelihood minus a penalty on the sum of absolute coefficient magnitudes. The tuning strength λ controls shrinkage, and can set some coefficients exactly to zero. Predictors should be standardized when penalties are otherwise scale-dependent. The intercept is normally unpenalized.

$$\hat{\beta} = \arg\min_{\beta} \{ -\ell(\beta) + \lambda \sum_{j=1}^{p} |\beta_j| \}$$

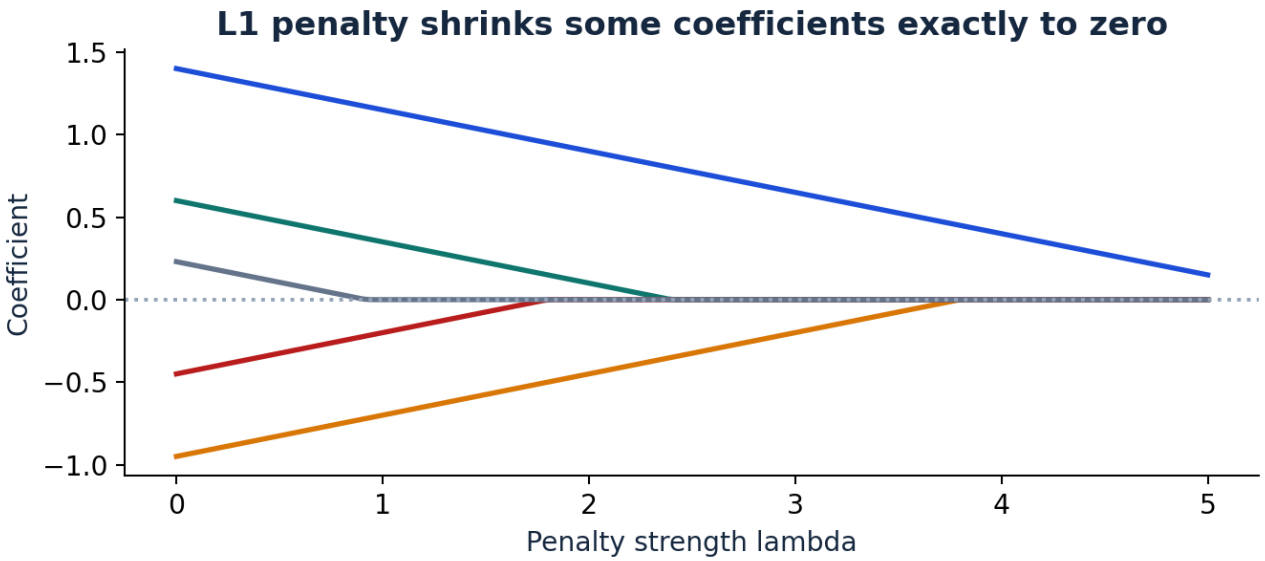


Figure. *Illustrative L1 coefficient paths. Stronger penalty reduces magnitude and can eliminate coefficients entirely; actual paths depend on the data.*

Lasso is useful when the number of predictors is large, but coefficient selection may be unstable with strong predictor correlation. The λ selected by cross-validation serves a predictive goal. Do not interpret all selected coefficients as independently established causal effects.

WORKED EXAMPLE

Question. Consider two standardized, highly correlated risk markers. What can happen when lasso is used?

Solution. The penalty can favor one marker and shrink the other to zero despite both carrying similar signal. Small sample changes can switch which marker is selected. For scientific interpretation, assess stability, biological plausibility, and confounding; for prediction, use honest external validation.

12. Discriminative logistic versus generative LDA

Chapter anchor: *ESL §4.4.5, printed pp. 127–128.*

Logistic regression directly models P(Y|X) and estimates slopes by conditional likelihood. LDA specifies class-conditional densities f(X|Y) and priors π, then obtains P(Y|X) via Bayes. Under shared normal covariance, both imply linear log-odds, but the parameter estimates differ because LDA imposes stronger distributional structure. If the structure is approximately true, LDA can be efficient; if badly wrong, flexibility can be preferable.

> **THE KEY DISTINCTION**
>
> **Same functional form does not imply same estimator.** LDA uses class means and covariances; logistic regression uses the conditional distribution of labels. Both can produce linear log-odds, but their likelihoods and assumptions differ.

> **MASTERY CHECK 9**
>
> If X|Y is visibly skewed and heavy-tailed, which distributional assumption is stressed more directly: logistic or Gaussian LDA? Why does that not guarantee logistic has better test performance?

13. Separating hyperplanes and Rosenblatt's perceptron

Chapter anchor: *ESL §4.5 and §4.5.1, printed pp. 129–132.*

For labels yi{−1,+1}, a hyperplane is the set x satisfying β0+β^{T}x=0. Its normal vector β is perpendicular to the boundary; the signed geometric distance from x to the hyperplane is (β0+β^{T}x)/||β||. A perceptron updates β when an observation is misclassified. For separable data, it can eventually find a separating hyperplane, but the result need not be the maximum-margin solution.

$$\text{distance}(x, \mathcal{H}) = \frac{\beta_0 + \beta^\top x}{\|\beta\|}$$

$$\beta \leftarrow \beta + \rho\, y_i x_i \quad \text{for a misclassified } i$$

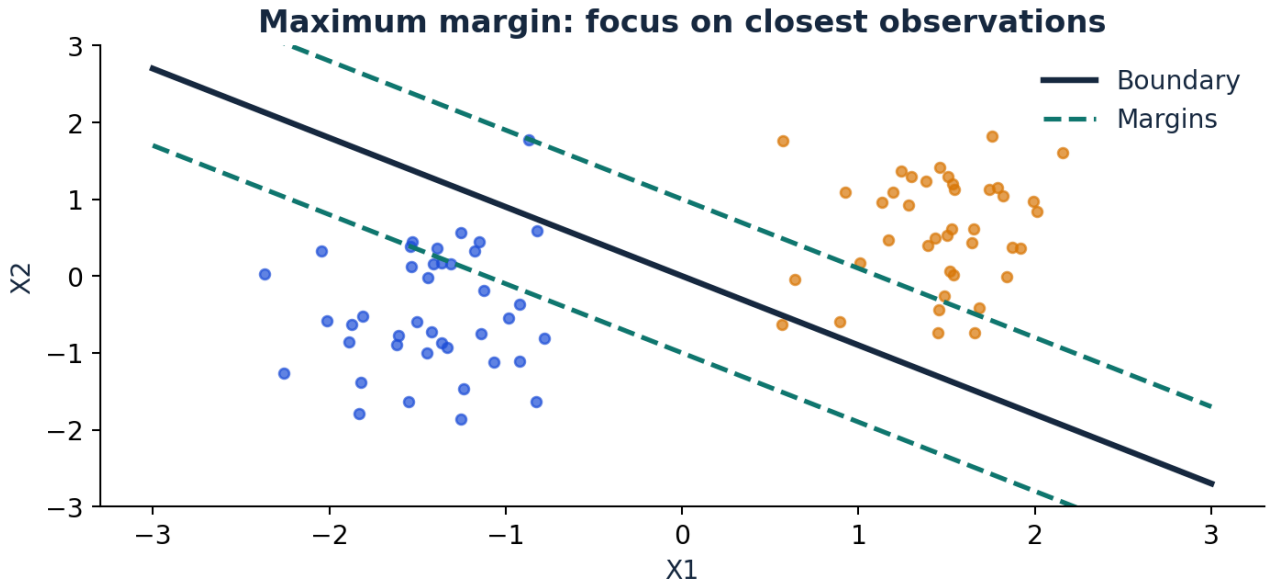


Figure. *Hyperplane orientation is controlled by its normal vector. The dashed lines depict a margin region around the decision boundary.*

WORKED EXAMPLE

Question. For β=(3,4), β0=−5 and x=(1,1), compute the signed geometric distance.

Solution. The score is 3+4−5=2, the norm ||β||=5, so the signed distance is 2/5=0.4. The sign identifies the side; distance normalizes out arbitrary scaling of the coefficient vector.

14. Optimal separating hyperplanes: maximum margin

Chapter anchor: *ESL §4.5.2, printed pp. 132–135. The supplied excerpt ends near the support-vector formulation; nonlinear kernels lie beyond this supplied chapter excerpt.*

Among all separating boundaries, the hard-margin support-vector formulation chooses the one with the largest minimum geometric distance to training observations. Rescale the coefficients so each correctly classified training point has signed score at least one. The margin width becomes 2/||β||, so maximizing it is equivalent to minimizing ||β||²/2 subject to the classification inequalities.

$$\min_{\beta,\,\beta_0} \tfrac{1}{2}\|\beta\|^2 \quad \text{s.t.} \quad y_i(\beta_0 + \beta^\top x_i) \geq 1$$

Using Lagrange multipliers αi≥0 yields β=Σαiyi xi. Only observations whose constraints are active can have positive αi; these are support vectors. The dual formulation makes dot products between observations central, preparing the transition to kernel methods in later chapters.

$$\beta = \sum_{i=1}^{n} \alpha_i y_i x_i, \qquad \alpha_i \geq 0$$

Hard margins assume perfect separability. Real clinical data overlap, labels can be noisy, and soft-margin extensions allow violations with a penalty. The source chapter excerpt here mainly establishes the separating-hyperplane geometry and optimal hard-margin optimization.

> **MASTERY CHECK 10**
>
> What is the central difference between a perceptron that finds *some* separator and the hard-margin support-vector formulation? Why can a separator that perfectly fits training data still be poor for prediction?

15. Theory synthesis: choosing a method

Logistic regression: direct conditional risk and interpretable odds ratios, provided the logit specification is reasonable. **LDA:** potentially stable under approximate Gaussian class distributions and shared covariance. **QDA:** curvature when class covariances truly differ, at a higher parameter cost. **Regularized discriminant:** middle ground between LDA and QDA. **Lasso:** prediction-oriented shrinkage and sparsity. **Support-vector methods:** margin-based classification emphasizing boundary observations rather than fitted probabilities.

> **RESEARCH RIGOR**
>
> Prefer prespecified predictors and sufficiently large validation data. Distinguish discrimination, calibration, and clinical decision usefulness. The birth-weight dataset used below is small, so it is a teaching example—not sufficient evidence for deployable clinical risk prediction.

PART II — APPLIED R LAB: NO LOOPS

The lab uses `MASS::birthwt` so that you can carry forward the dataset from your regression lessons. The outcome `low` is defined using `bwt`; therefore `bwt` must be removed from the predictors. Treat each numbered code block as one experiment. The companion .R file contains the same commands with no loops or custom functions.

Lab 1 • Data and leakage checks

An outcome derived from another column must not be predicted using that same deriving column. This is outcome leakage, even if the model appears excellent.

```
# Packages: install once if missing (not part of repeated analysis)
# install.packages(c("MASS", "Hmisc", "rms", "glmnet", "class"))
library(MASS)
library(Hmisc)
library(rms)

# MASS::birthwt: binary low birth weight is the outcome.
d <- MASS::birthwt
d$low <- factor(d$low, levels = c(0, 1), labels = c("No", "Yes"))
d$race <- factor(d$race, levels = c(1, 2, 3),
                 labels = c("White", "Black", "Other"))
d$smoke <- factor(d$smoke, levels = c(0, 1), labels = c("No", "Yes"))
d$ht <- factor(d$ht, levels = c(0, 1), labels = c("No", "Yes"))
d$ui <- factor(d$ui, levels = c(0, 1), labels = c("No", "Yes"))
d$ptl <- factor(d$ptl > 0, levels = c(FALSE, TRUE),
                labels = c("No", "Yes"))
d$bwt <- NULL       # CRITICAL: bwt defines low, so including it leaks outcome!
d$ftv <- NULL       # Omit unused variable to keep this lab focused.

# Inspect outcome frequency, missingness, and predictors.
Hmisc::describe(d)
table(d$low)
prop.table(table(d$low))
Hmisc::rcorr(as.matrix(d[, c("age", "lwt")]), type = "spearman")
```

```
# Stratified 70/30 split, preserving both outcome levels.
set.seed(2026)
train_no <- sample(which(d$low == "No"),
                   floor(0.70 * sum(d$low == "No")))
train_yes <- sample(which(d$low == "Yes"),
                    floor(0.70 * sum(d$low == "Yes")))
train_id <- sort(c(train_no, train_yes))
train <- d[train_id, ]
test <- d[-train_id, ]
table(train$low)
table(test$low)
```

Lab 2 • Linear probability versus logistic

Ask: Is the estimated line outside [0,1]? Where is a fitted probability changing most rapidly?

```
# Same predictor; compare probability ranges.
lpm <- lm(I(low == "Yes") ~ lwt, data = train)
logit <- glm(low ~ lwt, data = train, family = binomial())

summary(lpm)
summary(logit)
range(predict(lpm, newdata = test))
range(predict(logit, newdata = test, type = "response"))

# Draw both fitted curves, without a loop.
weight_grid <- data.frame(lwt = seq(min(train$lwt),
                                    max(train$lwt), length.out = 100))
plot(train$lwt, as.integer(train$low == "Yes"),
     pch = 16, cex = 0.6, xlab = "Maternal weight (lb)",
     ylab = "Low birth weight: 1=Yes, 0=No")
lines(weight_grid$lwt, predict(lpm, newdata = weight_grid),
      col = "orange", lwd = 2)
lines(weight_grid$lwt, predict(logit, newdata = weight_grid,
                               type = "response"),
      col = "blue", lwd = 2)
```

Lab 3 • Linear discriminant versus quadratic discriminant

On this small dataset, QDA may be noisier because it estimates one covariance per class. Compare held-out metrics rather than training classification alone.

```
# Use two numeric predictors for an interpretable first comparison.
lda_fit <- MASS::lda(low ~ age + lwt, data = train)
qda_fit <- MASS::qda(low ~ age + lwt, data = train)

lda_out <- predict(lda_fit, newdata = test)
qda_out <- predict(qda_fit, newdata = test)

# A posterior is class probability, not merely a hard class label.
head(lda_out$posterior)
head(qda_out$posterior)
table(Observed = test$low, Predicted = lda_out$class)
table(Observed = test$low, Predicted = qda_out$class)
mean(lda_out$class == test$low)
mean(qda_out$class == test$low)

# LDA coefficients define discriminant directions.
lda_fit$scaling
plot(lda_fit)
```

Lab 4 • rms logistic regression and conditional interpretation

Interpret the effect of maternal weight conditional on the other predictors. For probability contrasts, compute predictions at explicit covariate settings.

```
# datadist is needed for many rms prediction summaries.
dd <- rms::datadist(train)
options(datadist = "dd")

fit_rms <- rms::lrm(low ~ age + lwt + race + smoke + ptl,
                    data = train, x = TRUE, y = TRUE)
print(fit_rms)
coef(fit_rms)
exp(coef(fit_rms))       # odds ratios; intercept is not a group OR
rms::anova(fit_rms)     # effects, not a diagnostic residual plot
rms::vif(fit_rms)       # collinearity warning, not causal proof

p_rms <- plogis(predict(fit_rms, newdata = test, type = "lp"))
summary(p_rms)
Hmisc::somers2(p_rms, as.integer(test$low == "Yes"))
```

Lab 5 • Flexible effect with a restricted cubic spline

A nonlinear spline curve is not automatically better: examine uncertainty, plausibility, and internal validation.

```
# Three knots provide a modest nonlinearity allowance.
spline_fit <- rms::lrm(low ~ age + rcs(lwt, 3) + race +
                         smoke + ptl,
                       data = train, x = TRUE, y = TRUE)
print(spline_fit)
rms::anova(spline_fit)  # inspect the Nonlinear line for lwt
plot(rms::Predict(spline_fit, lwt, fun = plogis),
     xlab = "Maternal weight (lb)",
     ylab = "Predicted event probability")
```

Lab 6 • Likelihood, influence, and overfitting

The four plots screen for residual patterns and influential observations. They do not prove that the model is calibrated or causal.

```
# glm provides conventional generalized linear-model diagnostics.
fit_glm <- glm(low ~ age + lwt + race + smoke + ptl,
               data = train, family = binomial())
summary(fit_glm)
logLik(fit_glm)
AIC(fit_glm)

par(mfrow = c(2, 2))
plot(fit_glm, which = c(1, 2, 4, 5))
par(mfrow = c(1, 1))

# Refit/validate only with the training data.
set.seed(2026)
rms::validate(fit_rms, method = "boot", B = 100)
```

Lab 7 • L1-penalized logistic regression, no formula wrapper

The lambda search is trained only on the training partition. The held-out test set is not used to select lambda.

```
# model.matrix constructs numerical indicators for factor predictors.
x_train <- model.matrix(low ~ age + lwt + race + smoke + ptl,
                        data = train)[, -1]
x_test <- model.matrix(low ~ age + lwt + race + smoke + ptl,
                       data = test)[, -1]
y_train <- as.integer(train$low == "Yes")

set.seed(2026)
lasso <- glmnet::cv.glmnet(x_train, y_train, family = "binomial",
                           alpha = 1, nfolds = 5,
                           type.measure = "deviance",
                           standardize = TRUE)
plot(lasso)
coef(lasso, s = "lambda.min")
coef(lasso, s = "lambda.1se")

p_lasso <- as.numeric(predict(lasso, newx = x_test,
                              s = "lambda.min", type = "response"))
Hmisc::somers2(p_lasso, as.integer(test$low == "Yes"))
```

Lab 8 • Discrimination versus calibration

Check whether higher sensitivity is accompanied by additional false positives. The 0.20 cutoff is illustrative, not a recommended clinical threshold.

```
# Evaluate exactly the predictions made on the test data.
y_test <- as.integer(test$low == "Yes")
Hmisc::somers2(p_rms, y_test)  # C/AUC and Dxy
mean((y_test - p_rms)^2)       # Brier score; smaller is better

# Small samples make calibration plots quite noisy.
rms::val.prob(p_rms, y_test, pl = TRUE)

# Explicit 0.20 cutoff illustrates asymmetric clinical priorities.
pred_020 <- factor(ifelse(p_rms >= 0.20, "Yes", "No"),
                   levels = c("No", "Yes"))
table(Observed = test$low, Predicted = pred_020)
```

Lab 9 • Fisher discriminant directions and principal components

PCA maximizes total predictor variance. LDA directions aim to separate classes; neither is automatically causal.

```
# LDA yields a supervised linear projection.
lda_fit <- MASS::lda(low ~ age + lwt, data = train)
lda_fit$scaling
lda_fit$svd^2 / sum(lda_fit$svd^2)
plot(lda_fit)

# PCA does not use low birth weight class labels.
pca_fit <- prcomp(train[, c("age", "lwt")], scale. = TRUE)
summary(pca_fit)
pca_fit$rotation
plot(pca_fit$x[, 1], pca_fit$x[, 2],
     col = ifelse(train$low == "Yes", "orange", "blue"),
     xlab = "Unsupervised PC1", ylab = "Unsupervised PC2")
```

Interpretation and replication checklist

- Your event definition is exactly low birth weight (Yes), and the derivation variable bwt has been excluded.
- The same train/test rows are used when comparing methods; no threshold or penalty is selected from test performance.
- Odds ratios are not risk ratios. Treat discrimination (AUC), calibration (Brier/calibration plot), and decision cutoffs as different questions.
- The resulting estimates are illustrative due to the modest dataset size and possible covariate confounding.

Answer key — theory checks

1. $\beta=(0,0)$ produces a constant score with no predictor-dependent boundary. β_0 translates the boundary; changing β rotates its normal vector.

2. Posterior falls as π_1 decreases when the likelihood ratio is fixed. Prior prevalence is part of Bayes' numerator and denominator.

3. Polynomial features can yield nonlinear original-space boundaries, but limited basis choices can still misrepresent a complex multiclass structure.

4. $\exp(-0.7)\approx 0.497$: odds are multiplied by roughly one-half. This is not a fixed risk difference.

5. $p(1-p)$ is largest at $p=0.5$ (value 0.25) and approaches zero at the extremes. That drives the IRLS information weights.

6. Shared covariance terms cancel in a log ratio; class-specific quadratics generally remain, producing curved QDA boundaries.

7. Centered K class means span a space of dimension at most $K-1$, regardless of p; with 11 classes, at most 10 nonzero directions.

8. Softmax probabilities share a denominator; raising η_1 raises class 1 probability and decreases other class probabilities, holding their scores fixed.

9. Gaussian class-conditional LDA directly assumes normality; logistic does not, but logistic may still be misspecified or overfit.

10. The perceptron seeks any separating solution; max-margin optimization finds a separator with largest normalized gap. Neither guarantees generalization.

References and scientific scope

Primary source: uploaded *ESL Chapter 4*, *Linear Methods for Classification*, printed pp. 101–135, especially Equations 4.1–4.15 and 4.31–4.54. Figures in this lecture are newly generated conceptual illustrations, not copies of the book. Additional worked numbers, clinical framing, and the R laboratory are teaching extensions.

R documentation references: `?rms::lrm`, `?rms::validate`, `?rms::Predict`, `?Hmisc::somers2`, `?MASS::lda`, `?glmnet::cv.glmnet`. R code is written to be simple and reproducible but has not been executed in this authoring environment; check package versions and output before scientific use.